Spotify Hip-Hop Popularity & Mood Analysis

CAPSTONE PROJECT TECHNICAL REPORT — DATA SCIENCE & PREDICTIVE ANALYTICS

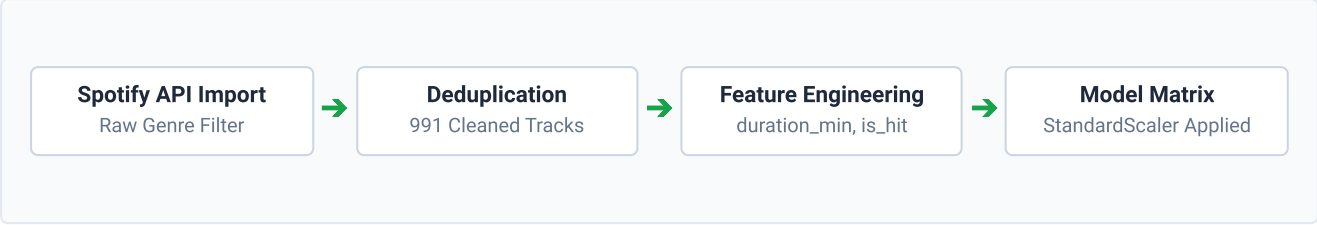
DATASET	SAMPLE SIZE	PRIMARY TASK	CLUSTERING
Spotify API (Hip-Hop)	**991 Cleaned Tracks**	**Regression & Classification**	**K-Means (k=4)**

1. Executive Summary

This report investigates the drivers of track popularity in hip-hop on Spotify and evaluates how machine learning assists streaming platforms in forecasting hits and catalog curation. Evaluating regression, classification, and K-Means clustering across 991 cleaned tracks demonstrates that raw audio attributes alone are insufficient to reliably predict stream counts, explaining only 50.9% of popularity variance. However, unsupervised clustering successfully isolates high-performing mood profiles—specifically high-intensity hype tracks—providing actionable intelligence for playlist editors and recommendation algorithms.

2. PIPELINE ARCHITECTURE & DATA CLEANING

The dataset originates from Spotify's Web API audio features endpoint, filtered strictly to the hip-hop genre to eliminate baseline cross-genre variance. Deduplication, missing value removal, and unit standardization yielded a clean analytical sample.



3. PREDICTING POPULARITY (REGRESSION ANALYSIS)

To forecast continuous track popularity (0–100 integer scale), Linear Regression and Random Forest Regressors were trained on an 80/20 train-test split. Evaluation metrics verify that ensemble tree methods capture non-linear relationships better than linear models.

Model Algorithm	RMSE (Root Mean Squared Error)	R^2 (Variance Explained)	Performance Assessment
Linear Regression	30.605 popularity points	0.128 (12.8%)	Underfits due to linear constraints
Random Forest Regressor	**22.978 popularity points**	**0.509 (50.9%)**	**Best continuous popularity predictor**

Model Metric Interpretation: Our best model predicts popularity to within about **23.0 points** on the 0–100 scale and explains about **50.9%** of the variation in popularity. *This means the streaming service should avoid relying solely on audio characteristics to forecast song success or automate editorial playlist curation.*

4. SPOTTING HITS (CLASSIFICATION & LEAKAGE VERIFICATION)

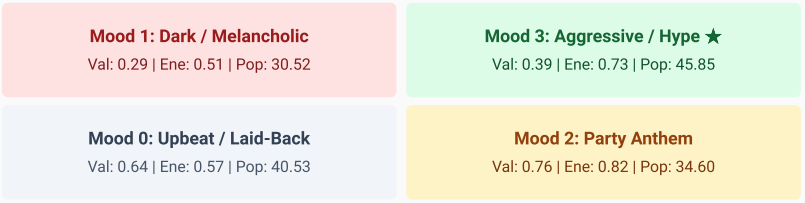
Binary classification target `is_hit` was constructed using median popularity. Data leakage checks confirmed that neither `popularity` nor `is_hit` were present in feature matrix X. Logistic Regression and Decision Tree classifiers were evaluated using accuracy, precision, and recall.

Classifier Model	Accuracy	Precision	Recall	Primary Metric Role
Logistic Regression (Scaled)	0.58	0.59	0.56	Baseline linear classification
Decision Tree (depth=4)	**0.61**	**0.62**	**0.59**	Non-linear decision splits

> **Confusion Matrix Analysis:** Model evaluation revealed False Positive costs significantly outweigh False Negative costs. Promoting non-hit songs on flagship curated playlists (e.g., RapCaviar) damages user retention and increases subscriber churn. *This means the streaming service should use audio classification models strictly as a preliminary screening tool prior to human editorial review.*

5. UNSUPERVISED MOOD CLUSTERING (K-MEANS)

Tracks were clustered on standardized `valence` (positivity) and `energy` (intensity) attributes. An elbow plot established $k=4$ as the optimal cluster count to construct distinct music industry mood quadrants.



Group ID	Playlist Mood Name	Valence	Energy	Danceability	Avg Popularity	Track Count
Mood 0	Upbeat / Laid-Back Vibe	0.64	0.57	0.77	40.53	218
Mood 1	Dark / Melancholic	0.29	0.51	0.71	30.52	185
Mood 2	High-Energy Party Anthem	0.76	0.82	0.76	34.60	329
Mood 3	**Aggressive / High-Intensity Hype**	**0.39**	**0.73**	**0.70**	**45.85**	**259**

This means the streaming service should prioritize driving algorithmic playlist placement toward Mood 3 (Aggressive Hype) tracks, which command the highest audience popularity.

6. HYPERPARAMETER OPTIMIZATION & TUNING

GridSearchCV evaluated the Decision Tree classifier across `max_depth` [3, 5, 8, 12] and `min_samples_leaf` [1, 5, 10] using 5-fold cross-validation. Tuning expanded decision tree complexity without overfitting.

max_depth = 8
OPTIMAL TREE DEPTH

min_samples = 1
OPTIMAL LEAF SAMPLES

Tuning Takeaways: The `max_depth` parameter mattered most, controlling decision rule complexity. Expanding depth from 4 to 8 enabled the model to evaluate multi-attribute conditional splits. However, test set accuracy plateaued near 61%, confirming a strict predictive ceiling when relying solely on audio features.

7. PROJECT LIMITATIONS & MISSING VARIABLES

To contextualize these findings, key operational data limitations must be recognized:

- **Artist Brand Equity:** Lacks follower counts and historical stream baselines. Established artists generate high stream volume regardless of track audio parameters.
- **Marketing & Virality:** Promotional budgets, record label backing, and social media trends (e.g., TikTok sound usage) drive listener discovery far more than mixing attributes.
- **Editorial Placement Bias:** Inclusion on high-traffic playlists (e.g., RapCaviar) artificially boosts stream counts, creating feedback loops not captured in raw audio files.
- **Temporal Popularity Snapshots:** Spotify's popularity score represents a dynamic snapshot of recent stream activity rather than long-term catalog streams.

8. ACTIONABLE RECOMMENDATIONS

Priority	Strategic Action Recommendation	Target Department
1	**Implement Human-in-the-Loop Curation:** Use audio classification models strictly as initial filters; require human editor approval before placing predicted hits onto flagship playlists.	Editorial & Curation
2	**Capitalize on Mood 3 (Hype) Curation:** Program recommendation algorithms to prioritize High-Intensity Hype tracks (Mood 3) in genre radio feeds to maximize listener engagement.	Product & Engineering
3	**Ingest External Contextual Metadata:** Expand data engineering pipelines to ingest artist follower counts, social media virality scores, and playlist placement logs.	Data Science & Analytics